

- **Problem Statement ID – SIH26120**
- **Problem Statement Title-** Digital Twin for Well-to-Surface Optimisation of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.
- **Theme- Smart Automation**
- **Department- Oil India Limited**
- **PS Category- Software**
- **Team Name- Fluxora**

Solution**Technical Approach****Feasibility & Viability****Impact & Benefits**



Proposed Solution

A closed-loop digital twin that continuously links reservoir thermal state to artificial-lift control - replacing today's disconnected, manual, experience-based tuning of CSS and SRP operations

Salient Features

Coupled Subsurface-to-Surface Integration

Dynamically links post-injection reservoir cooling and fluid viscosity decay directly to real-time Sucker Rod Pump (SRP) speed control.



Physics-Informed Machine Learning Architecture

Combines physical heat-loss and pump kinematic laws with AI models to guarantee physically realistic, safe predictions.



Multi-Objective Model Predictive Control (MPC)

Evaluates future CSS and SRP settings over a rolling horizon to maximize crude recovery while minimizing steam usage, electricity costs, and structural risks.



Predictive Mechanical Reliability & Anomaly Detection

Uses high-frequency dynamometer card analysis to detect rod-floating anomalies early and prevent mechanical pump failures.



Continuous State Estimation & Explainable AI

Estimates unobservable subsurface states in real time while delivering transparent, physics-backed rationale for every system recommendation.



What's Unique?

Truly Coupled

Reservoir thermal prediction feeds directly into SRP setpoint recommendations, not two isolated tools



Predictive, Not Reactive

Rod floating and Impact loading are flagged before failure, from live dynamometer / VFD signal patterns

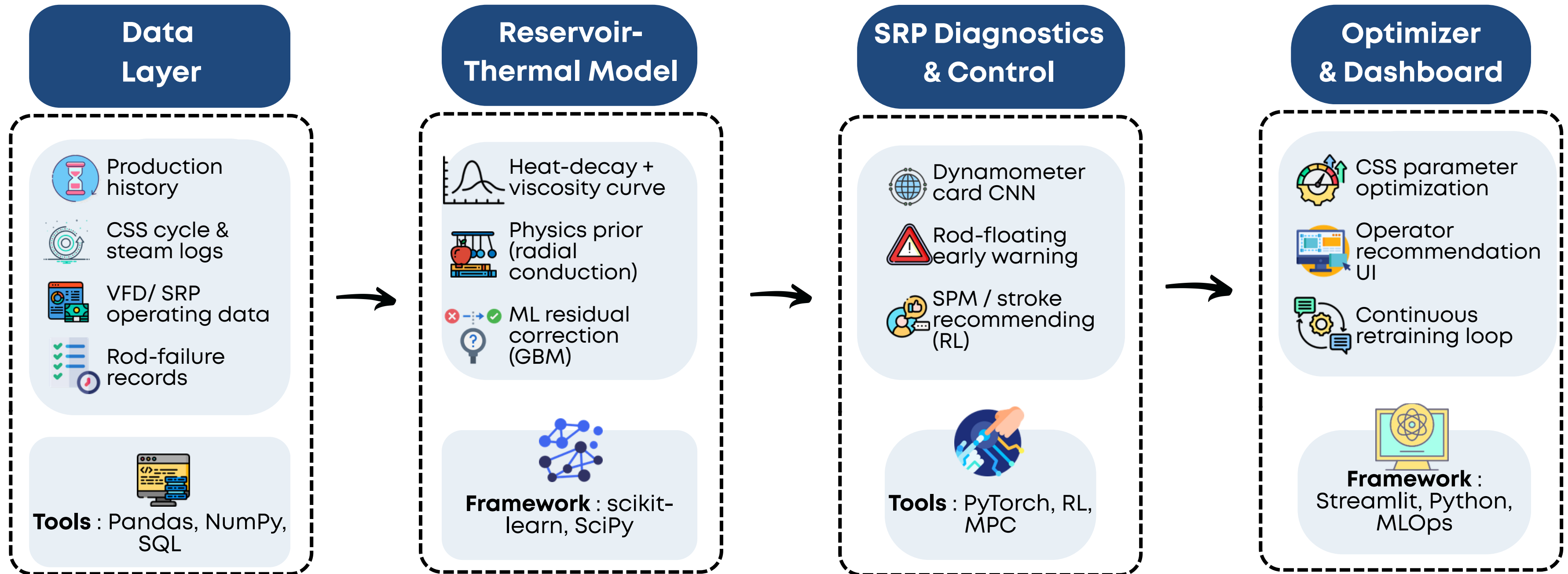


Built for Field Reality

Hybrid physics + ML design works with the sparse, noisy data typical of onshore heavy-oil assets



Technical Approach





Feasibility & Viability

Feasibility



• **Zero Hardware Need:** Operates purely as software using existing field logs (CSS, VFD, failures).



• **Rapid Prototyping:** Prototype-ready immediately using analog production datasets.



• **Proven Foundation:** Leverages literature-validated thermal and pump kinematics models.

Viability



• **Phased Rollout:** Advisory-first deployment before advancing to closed-loop automation.



• **High ROI / Zero CAPEX:** Drives immediate OPEX savings (steam, energy) without new hardware.



• **Low Technical Risk:** Combines proven physical models rather than unproven ML.

Challenges



• **Data Fragmentation:** Solved by building a unified per-well, time-indexed schema early on.

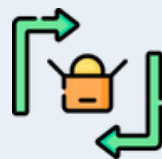


• **Rare Failure Data:** Tackled using physics-informed features and anomaly detection framing.



• **Legacy SCADA Risk:** Managed via advisory-first deployment prior to automated field control.

Future Scope



• **Closed-Loop Automation:** Direct automated setpoint adjustments via field SCADA protocols.

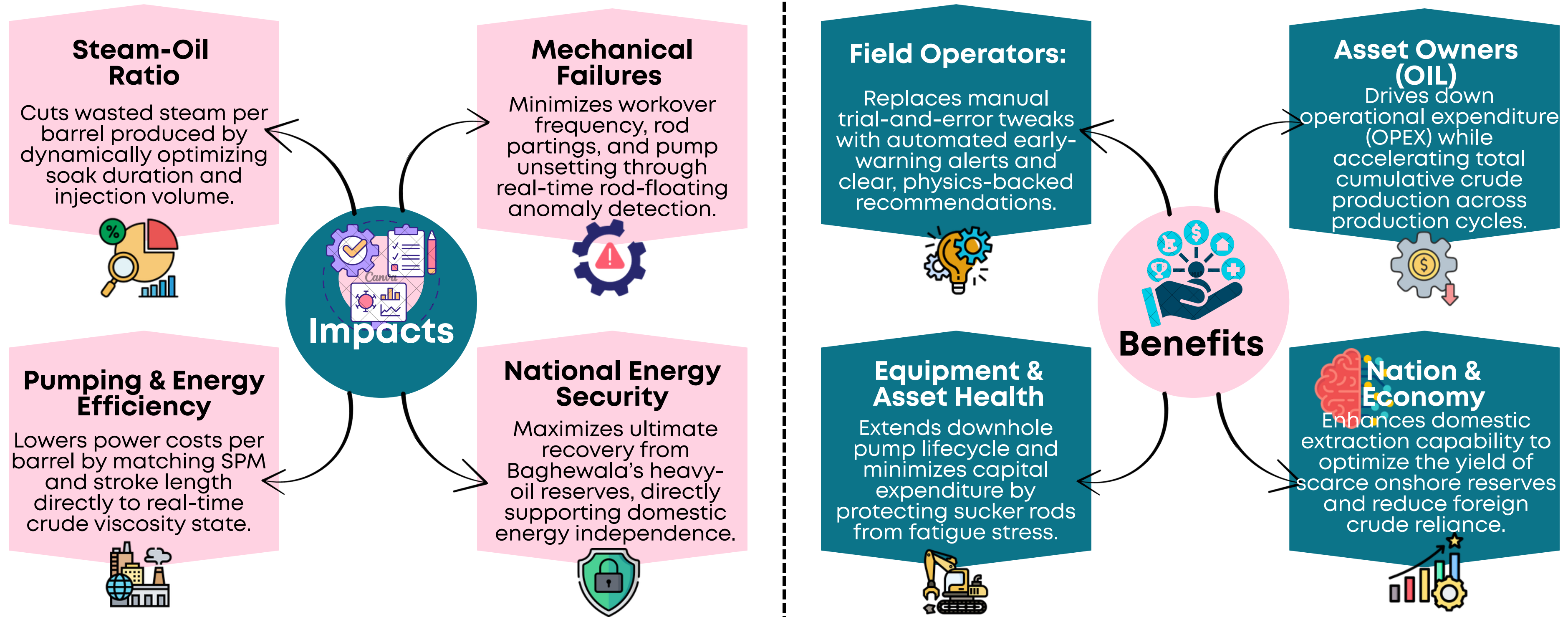


• **Pad-Level Expansion:** Scaling from single wells to pad-wide steam allocation networks.



• **Cross-Lift Adaptation:** Extending the thermal-viscosity engine to Progressive Cavity Pumps.

Impact & Benefits



CSS Thermal & Reservoir Optimization



- Zhao, L. et al. (2021). An Investigation of Production Performance by Cyclic Steam Stimulation in Heavy Oil Reservoirs. Energy
- Li, D. et al. (2023). Machine-Learning-Assisted Identification of Steam Channeling After Cyclic Steam Stimulation in Heavy-Oil Reservoirs. Energy & Fuels
- SPE (2019). Simulation-Calibrated Machine Learning for Cyclic Steam Production Forecasting. Society of Petroleum Engineers
- Montgomery, D. C. (2017). Box-Behnken Response-Surface Optimization of CSS Operational Parameters. Design and Analysis of Experiments

SRP Diagnostics & Intelligent Control



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Digital Twin & Autonomous Decision-Making



- Rebello, S. et al. (2023). Digital Twin Framework for Optimal and Autonomous Decision-Making in Cyber-Physical Systems
- Liu, Y. et al. (2024). Digital-Twin Control Framework for Beam Pumping Units and Rod Fatigue Mitigation
- Networked Digital SRP Systems for Continuous Automated Pump-Off Optimization

Physics-Informed ML & Optimization



- Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-Informed Neural Networks: A Deep Learning Framework for Physical Systems
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Solution

Technical Approach

Feasibility & Viability

Impact & Benefits